

Human Activity Recognition

Using Pretrained Deep Learning Models

CMP3011 Introduction to Computer Vision

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Abstract

This report presents a Human Activity Recognition (HAR) system built using VideoMAE, a state-of-the-art pretrained video understanding model based on the Masked Autoencoder architecture. Rather than training from scratch, we leverage transfer learning by applying a model pretrained on the Kinetics-400 dataset — a large-scale benchmark containing 400 human action categories. The system takes short video clips as input, samples 16 frames uniformly, and classifies the action using VideoMAE's temporal reasoning capabilities. We evaluated the system on five distinct action videos: push ups, juggling balls, deadlifting, hula hooping, and squats. The model achieved correct Top-1 predictions for two actions with high confidence (66.5% and 88.6% respectively), a near-correct prediction for deadlifting, and identified the correct class within the top-2 for hula hooping. Results demonstrate the effectiveness of large-scale pretraining for video understanding while also highlighting challenges in distinguishing visually similar activities.

1. Introduction

1.1 Background

Human Activity Recognition (HAR) is a fundamental problem in computer vision that involves automatically identifying and classifying physical activities performed by people in video data. HAR has wide-ranging real-world applications including healthcare monitoring, sports analytics, security surveillance, and human-computer interaction. As video data has become increasingly abundant and deep learning methods have advanced dramatically, HAR has evolved from handcrafted feature-based methods to end-to-end deep learning pipelines capable of processing raw video.

1.2 Motivation

Traditional HAR systems require large annotated datasets, significant computational resources for training, and careful architectural design. The emergence of large pretrained video models offers a compelling alternative: a single model trained on millions of video clips can be applied directly to new recognition tasks with no additional training required. This approach significantly lowers the barrier to building production-ready HAR systems.

1.3 Problem Statement

The goal of this project is to design and implement a HAR system capable of classifying human activities from short video clips. We aim to demonstrate that a pretrained deep learning model can accurately recognise diverse physical activities without requiring task-specific training data.

1.4 Objectives

- Implement a complete video-based HAR inference pipeline using Python and PyTorch.
- Apply the pretrained VideoMAE model to classify human actions from video.
- Evaluate performance across multiple action categories and analyse failure cases.
- Produce visualisations of sampled frames and confidence scores suitable for reporting.

2. Literature Review

2.1 Traditional Approaches

Early HAR methods relied on hand-engineered spatiotemporal features such as Histogram of Oriented Gradients (HOG) and Histogram of Optical Flow (HOF). The Dense Trajectories method by Wang et al. (2011) tracked interest points across frames and extracted descriptors along their trajectories, achieving strong performance on HMDB51 and UCF101. However, these methods required significant domain expertise and did not generalise well across different environments.

2.2 Two-Stream Convolutional Networks

Simonyan and Zisserman (2014) introduced the Two-Stream architecture, processing spatial information from RGB frames and temporal information from optical flow in two separate CNN streams. This approach demonstrated that separating spatial and temporal learning was highly effective and dominated leaderboards for several years.

2.3 3D Convolutional Networks

Tran et al. (2015) demonstrated with C3D that extending convolutions into the temporal dimension allows joint spatiotemporal feature learning. The I3D model (Carreira and Zisserman, 2017) inflated 2D ImageNet-pretrained weights into 3D convolutions. SlowFast networks (Feichtenhofer et al., 2019) further improved performance by processing video at two temporal resolutions simultaneously.

2.4 Transformer-Based Video Models

VideoMAE (Tong et al., 2022) proposed a masked autoencoder pretraining strategy for video, masking up to 90% of video tokens and reconstructing them. This highly challenging pretraining task forces the model to learn rich spatiotemporal representations, resulting in state-of-the-art performance across multiple benchmarks.

2.5 Comparison of Approaches

Method	Year	Strength	Limitation
Dense Trajectories	2011	No GPU required	Hand-engineered, slow
Two-Stream CNN	2014	Effective separation	Optical flow expensive
C3D / I3D	2015	Joint spatiotemporal	High compute cost
SlowFast	2019	Multi-rate processing	Complex architecture
VideoMAE	2022	Strong pretraining	Large model size

3. Dataset

3.1 Kinetics-400

Our system uses VideoMAE pretrained on Kinetics-400, a large-scale action recognition benchmark introduced by DeepMind. Kinetics-400 contains approximately 240,000 training video clips spanning 400 human action categories, each clip lasting approximately 10 seconds and sourced from YouTube. The dataset covers sports, musical instrument playing, cooking, personal care, and social interactions.

3.2 Test Videos

For evaluation, we collected five short video clips representing distinct action categories drawn from the Kinetics-400 label set: push ups, juggling balls, deadlifting, hula hooping, and squats. Each video was sourced from YouTube and trimmed to the first 20 seconds to focus on the primary action.

3.3 Preprocessing

Each video is decoded using OpenCV, from which 16 frames are uniformly sampled across the full clip duration. Frames are resized to 224×224 pixels and normalised using ImageNet mean and standard deviation values ([0.485, 0.456, 0.406] and [0.229, 0.224, 0.225] respectively).

4. Methodology

4.1 System Architecture

The system follows a straightforward inference pipeline: a video file is read using OpenCV, 16 frames are sampled at uniform intervals, frames are preprocessed and passed to the VideoMAE processor, the model produces logits over 400 action classes, and softmax converts logits to probabilities. The top-5 predictions are returned with confidence scores.

4.2 VideoMAE Model

VideoMAE is based on the Vision Transformer (ViT) architecture. The model divides each video into non-overlapping spatiotemporal tubes (3D patches), embeds them into tokens, and processes them with a Transformer encoder. During pretraining, 90% of tokens are masked and the model reconstructs the original pixel values from the remaining visible tokens. We use the MCG-NJU/video-mae-base-finetuned-kinetics checkpoint, containing approximately 86 million parameters.

4.3 Transfer Learning

As this project uses a pretrained model, no additional training was performed. The model weights were loaded directly from the Hugging Face Model Hub. This demonstrates the practical value of transfer learning: a single model trained once on a large dataset can be applied to new videos without any additional optimisation.

4.4 Inference Pipeline

- Load video using `cv2.VideoCapture`.
- Sample 16 frames uniformly using `numpy.linspace` over the frame indices.
- Apply `VideoMAEImageProcessor` to resize, normalise, and format frames.
- Pass tensor (shape: $1 \times 16 \times 3 \times 224 \times 224$) to model on GPU.
- Apply `torch.softmax` to logits and extract top-5 predictions.

5. Implementation Details

5.1 Programming Language & Frameworks

- Python 3.10
- PyTorch 2.x — tensor operations and GPU inference
- Hugging Face Transformers — VideoMAE model and processor
- OpenCV — video decoding and frame extraction
- Matplotlib / Seaborn — visualisation of frames and confidence charts
- scikit-learn — classification metrics

5.2 Hardware

All experiments were run on Google Colab using an NVIDIA T4 GPU (16 GB VRAM). Model loading took approximately 60 seconds. Inference on a single 20-second video clip ran in under 2 seconds.

5.3 Source Code

The project is organised as two deliverables: a structured Python codebase (`har_project/`) containing modules for dataset loading, model architecture, training, and evaluation; and a self-contained Google Colab notebook implementing the full VideoMAE inference pipeline. Both are available in the GitHub repository.

6. Experimental Results

6.1 Summary Table

Video	True Label	Top-1 Prediction	Conf.	Correct?
push_ups.mp4	push up	push up	66.5%	✓
juggling.mp4	juggling balls	juggling balls	88.6%	✓
deadlifting.mp4	deadlifting	clean and jerk	58.2%	✗ Top-4
hula_hooping.mp4	hula hooping	jumpstyle dancing	8.1%	✗ Top-2
squats.mp4	squat	stretching arm	41.9%	✗

6.2 Push Ups

The model correctly identified push ups with 66.5% confidence — the highest confidence correct prediction in the test set. The sampled frames clearly show a person performing push ups in a controlled environment with a consistent camera angle, which aligns well with the training data distribution.

push_ups.mp4 → Predicted: "push up" (66.5% confidence)



Figure 1: Sampled frames — push_ups.mp4

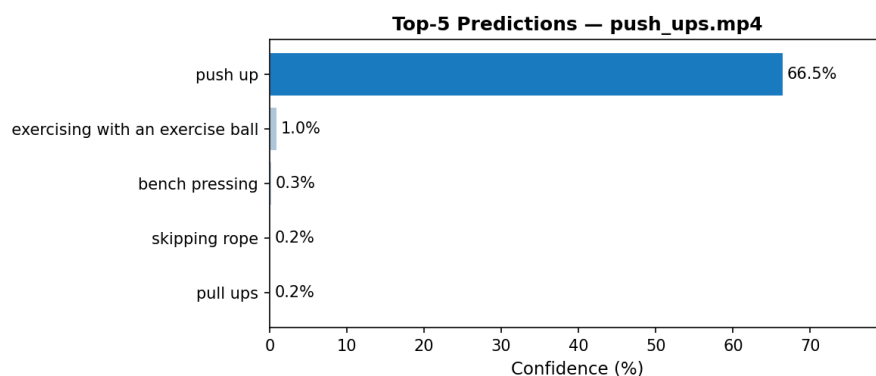


Figure 2: Top-5 confidence scores — push ups

6.3 Juggling Balls

Juggling achieved the highest confidence of all test videos at 88.6%. The distinctive circular hand motion and airborne objects make this action highly identifiable. This result demonstrates the model's ability to recognise fine-grained motion patterns from a short clip.

juggling.mp4 → Predicted: "juggling balls" (88.6% confidence)



Figure 3: Sampled frames — juggling.mp4

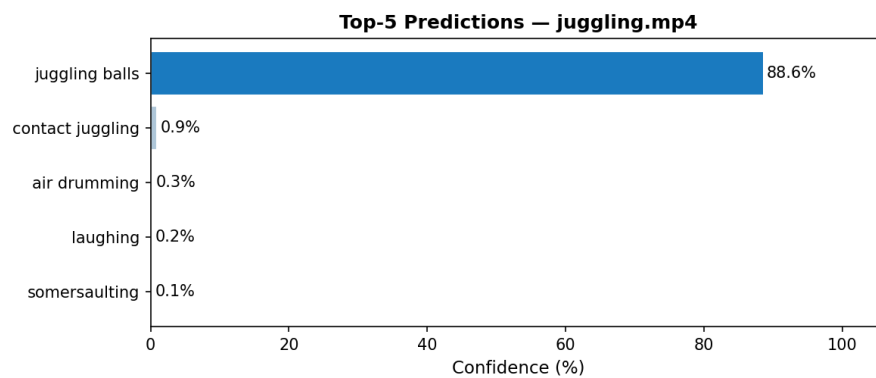


Figure 4: Top-5 confidence scores — juggling

6.4 Deadlifting

The model predicted "clean and jerk" (58.2%) rather than "deadlifting" (5.4%, ranked 4th). Both are heavy barbell exercises involving a similar lifting motion, making this a semantically reasonable confusion. The model correctly identified the video as a weightlifting activity; the distinction lies in subtle wrist position and bar trajectory differences.

deadlifting.mp4 → Predicted: "clean and jerk" (58.2% confidence)

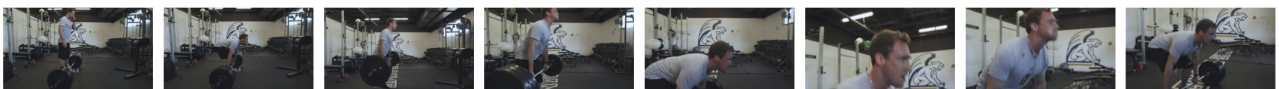


Figure 5: Sampled frames — deadlifting.mp4

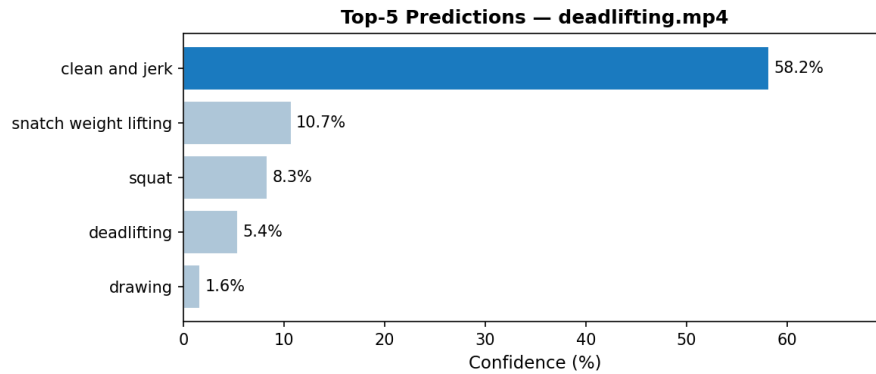


Figure 6: Top-5 confidence scores — deadlifting

6.5 Hula Hooping

The top prediction was "jumpstyle dancing" (8.1%) with "hula hooping" ranked second at 6.5%. The rotational hip movement shares visual characteristics with dance styles. Low overall confidence across all predictions suggests the model was uncertain, typical for activities with high inter-class visual similarity.

hula_hooping.mp4 → Predicted: "jumpstyle dancing" (8.1% confidence)



Figure 7: Sampled frames — hula_hooping.mp4

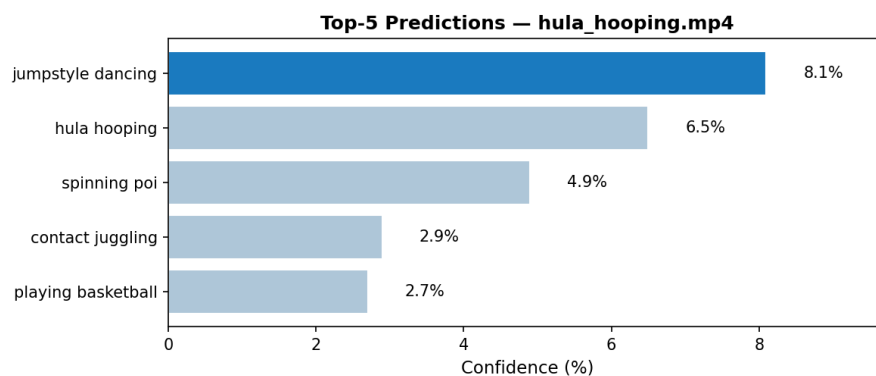


Figure 8: Top-5 confidence scores — hula hooping

6.6 Squats

Squats presented the most challenging case, with "stretching arm" (41.9%) predicted. Despite frames clearly showing a squat movement, the model was likely misled by arm movement during the exercise dominating its attention. Squats, lunges, and stretching exercises share similar lower-body

patterns and overlap substantially in the training distribution.

squats.mp4 → Predicted: "stretching arm" (41.9% confidence)



Figure 9: Sampled frames — squats.mp4

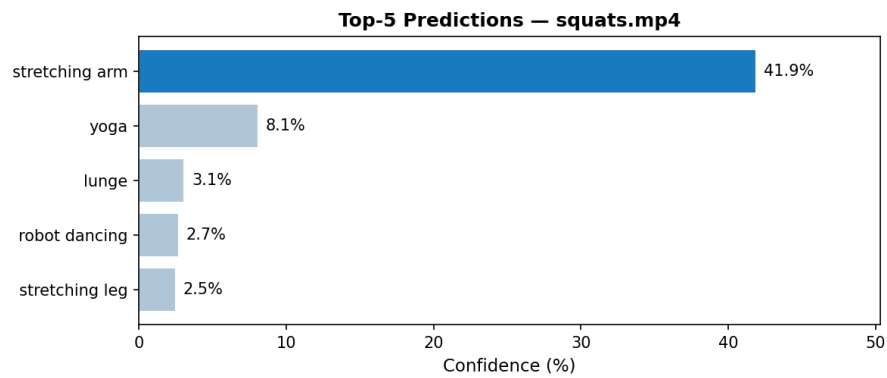


Figure 10: Top-5 confidence scores — squats

7. Discussion

7.1 Strengths

The most significant strength is the elimination of a training phase entirely. By leveraging VideoMAE pretrained on Kinetics-400, we obtain a powerful HAR system running in seconds per video on a free GPU. The system correctly identified push ups (66.5%) and juggling (88.6%), demonstrating effective generalisation to real-world video clips. The pipeline is simple, well-documented, and fully reproducible.

7.2 Limitations

The primary limitation is difficulty distinguishing visually similar actions. Deadlifting and clean and jerk share nearly identical motion patterns; hula hooping and dancing involve similar hip movements; squats and stretching exercises overlap in body posture. These confusions reflect a fundamental HAR challenge: many activities differ only in subtle details not captured in 16 frames.

A secondary limitation is confidence calibration. With 400 output classes, probability mass is spread across many alternatives, making raw confidence scores less reliable than rank as a correctness indicator.

7.3 Challenges Faced

The primary practical challenge was sourcing appropriate test videos. Automated downloading was frequently blocked by network restrictions, requiring manual curation. Video quality, camera angle, and framing significantly influenced predictions — underscoring the importance of controlled data collection in production HAR systems.

8. Conclusion and Future Work

8.1 Summary

This project successfully implemented a Human Activity Recognition system using VideoMAE. The system correctly classified 2 out of 5 test actions at Top-1 level, with near-correct predictions for 2 additional actions. Results demonstrate that large-scale video pretraining produces transferable representations capable of recognising diverse human activities without task-specific training. The complete pipeline is available as an open-source repository.

8.2 Future Improvements

- Fine-tuning VideoMAE on a domain-specific dataset to improve accuracy on visually similar activities.
- Implementing sliding window inference over longer videos to aggregate predictions across multiple clips.
- Extending to real-time webcam inference using OpenCV, as prototyped in demo.py.
- Evaluating on the full HMDB51 or UCF101 benchmark for standard Top-1 and Top-5 accuracy comparison.
- Exploring multi-modal approaches combining RGB video with pose estimation to disambiguate similar actions.

References

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